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AI-Powered IoT Framework for Real-Time Sleep Quality Monitoring and Enhancement

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***Abstract*—Sleep is very important for keeping both physical and mental health, but it is often ignored in everyday life. Like eating well and exercising regularly, getting enough sleep is important for keeping your health in general. Over the past few years, quick changes in technology and hard work schedules have made sleep more irregular and disturbed the habits. Not getting enough sleep or getting bad sleep has been linked to several detrimental health conditions, including obesity and metabolic disorders, diabetes and heart problems. This work is the result of a thorough analysis of the present problems with sleep quality, the requirement for ongoing sleep monitoring, usage of the Internet of Things (IoT) technologies and the open research questions in this field. This paper's goal is to evaluate sleep quality and promote enhancement by utilizing cutting-edge technology advancements that incorporate intelligent data-driven techniques. The framework uses IoT tools to collect both the physical and environmental data, which is then processed using the Machine Learning techniques (ML) to identify the factors that reduces the quality of sleep. Compared to traditional sleep monitoring methods like polysomnography (PSG), which are accurate but costly, time-consuming and unsuitable for continuous at-home use, the Internet of Things (IoT) system powered by artificial intelligence (AI) is less expensive and collects data with less human intervention. Experimental evaluation shows the system's potential for continuous monitoring and accurate sleep pattern recognition.**

Keywords*—Sleep Quality, Health, Artificial Intelligence (AI), Internet of Things (IoT), traditional methods and Machine Learning.

I. INTRODUCTION

Maintaining overall health requires getting enough sleep. It helps the body and brain recover from the events of the day by acting as the body's natural reset button. For most individuals, getting between seven to nine hours of sleep each night is essential to maintain the optimal functioning. Your mood, immunity, memory, emotional stability and ability to make better judgments all improve when you get adequate sleep. It is also vital for a healthy heart and keeping your metabolism in check over time. When you don't get enough sleep regularly, it can lead to various problems, both physical and mental. Things like high blood pressure, weight gain, anxiety and feeling down are more likely are some of the symptoms of not getting enough sleep. Plus, being constantly tired messes with your ability to perform well daily and generally lowers your quality of life [2].

For a significant portion of the population, sleep rhythms are disrupted by the widespread use of smartphones and other highly engrossing digital devices. People may experience several major problems if they don't get enough sleep. Early detection and ongoing sleep pattern monitoring are necessary to prevent the long-term health problems [7], [9]. Traditional diagnostic methods such as polysomnography (PSG), are still the clinical gold standard for sleep assessment. However, PSG requires special equipment, hospital stays and a human supervision, which makes it costly and impractical for long-term use. Recently, the use of Internet of Things (IoT) devices combined with Artificial Intelligence (AI) has become a promising alternative for continuous real-time sleep monitoring. The development of the Internet of Things (IoT) technology helps us to build a consistent and cost-effective system to monitor the sleep quality. IoT has both wearable and non-wearable sensors which helps to collect the physical data including heart-rate, oxygen levels and environmental data such as temperature, humidity and noise [1], [4].

Machine learning and deep learning techniques have demonstrated strong potential in extracting meaningful patterns from large-scale physiological datasets. Algorithms such as Random Forest, Support Vector Machines and deep neural networks can effectively model non-linear relationships in sleep-related data, providing accurate classification and prediction capabilities. Moreover, AI-based personalization allows systems to adapt to individual sleep behaviours over time, improving long-term accuracy and user relevance [5], [8], [11].

In addition, recent advancements in cloud and edge computing architectures have enhanced the scalability and responsiveness of IoT-based healthcare systems. Cloud-integrated AI models support continuous health data streams and long-term trend analysis, while edge computing enables low-latency processing for real-time feedback. These technological developments are particularly relevant for sleep monitoring applications that require both immediate intervention and historical sleep analysis [12], [13].

Despite these advancements, existing sleep monitoring frameworks often focus solely on data collection or offline analysis, lacking an integrated approach for real-time monitoring, intelligent analysis and sleep quality enhancement.

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This research aims to address these limitations by proposing an AI-powered IoT framework that combines continuous sensing, intelligent analytics and adaptive feedback mechanisms to support personalized sleep management in real-world environments [1], [3], [15].

II. Existing System

Existing sleep monitoring systems primarily rely on conventional clinical and consumer-grade approaches that offer limited adaptability and scalability. Polysomnography (PSG) is widely considered the gold standard for sleep analysis, as it records all kinds of the physiological signals commonly known as electroencephalogram (EEG), electrocardiogram (ECG), eye movement and muscle activity. Although PSG provides highly accurate sleep stage classification, it also requires specialized equipment, trained personnel and the controlled laboratory environments, making it costly and unsuitable for continuous long-term monitoring in real-world conditions [1], [2].

In recent years, wearable-based sleep tracking devices have gained popularity due to their portability and ease of use. These devices utilize embedded sensors to capture physiological parameters such as heart rate, motion and respiration. While wearables only enable continuous monitoring in the home environments, their accuracy varies significantly across devices and is often limited by sensor noise and simplified sleep scoring algorithms. Additionally, many wearable systems mostly depend on proprietary algorithms, limiting transparency and adaptability for personalized healthcare applications [3], [4]. Wearable technology, including fitness bands, smartwatches and chest straps, has been used in recent years to track sleep using accelerometer data and simple heart rate measures. Despite being handy, these devices usually have poor accuracy and are unable to distinguish between different stages of sleep and can lead to measurement errors because of motion distortions [7].

Physiological signals are gathered and sent to cloud servers for processing in some IoT-enabled systems that have been proposed for remote health monitoring. However, these systems encounter a number of difficulties, such as:

- High Latency: Processing of the data and its transformation delays in the real-time monitoring using the ancient and complexed methods.
- Limited Intelligence: These systems are not under process for using and combining the less complexity algorithms for the analysis of sleep stage classification.
- Scalability Issues: The existing frameworks are expensive, specialized sensors limits accessibility and large-scale employment [4], [6], [9].

Existing AI-based solutions have demonstrated potential in pattern recognition and anomaly detection, but most lack integration with IoT for real-time, continuous and adaptive sleep quality management. Consequently, current systems either focus solely on monitoring are constrained by the hardware limitations, leaving a research gap in end-to-end intelligent frameworks that combine monitoring with active sleep quality improvement [1], [3], [9].

III. PROPOSED SYSTEM

A. *System Overview:*

The suggested system is an AI-powered IoT solution for continuous monitoring and improvement of sleep quality in a non-invasive manner. Unlike typical sleep monitoring methods, which completely aim to study or monitor how someone sleeps, this proposed approach creates a comprehensive and qualitative way by incorporating the Internet of Things (IoT) for sensing and Artificial intelligence for the analytics process. As a part of this, the Random forest Algorithm is used for better prediction, handling the feature importance and noisy data [1], [4].

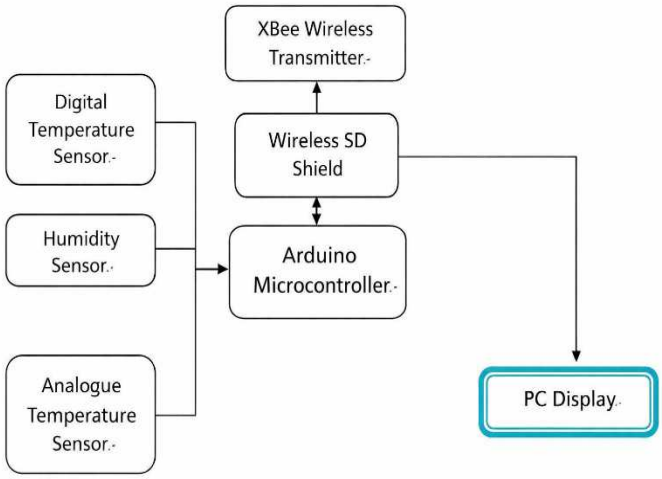


Fig. 1. ML-Based Work flow for Sleep Monitoring

The system integrates Internet of Things (IoT) sensors and devices to collect physiological and environmental measures; will input them to biosensor data, environmental context and potentially prior night's sleep data, to the machine learning models for sleep stage classification, evaluation of sleep quality and to provide real time actions to optimize sleep outcomes Fig(1). Machine learning and deep learning algorithms are employed to analyse sleep-related data and classify sleep stages accurately. By learning individual sleep patterns over time, the AI models adapt to user-specific characteristics, enabling the personalized sleep quality assessment and anomaly detection. This adaptive learning capability enhances prediction accuracy and supports early identification of the potential sleep disorders [5], [11].

B. *Key Components:*

- IoT-Based Sensors: These are used collect physiological data (such as heart rate, oxygen and body movement) and environmental data (including temperature, humidity, light and noise). Wireless communication modules (Wi-Fi/Bluetooth) are used to safely transfer data.
- Processing: The data preprocessing includes feature extraction, noise removal, normalization and executes the machine learning algorithm which is the Random

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- Forest Model to define sleep stages, evaluate the quality of sleep and identify any kind of sleep disorders (such as insomnia and sleep-apnea).
- System Workflow:
- Data Acquisition: IoT-based sensors continuously record environmental factors like temperature, humidity, light and noise as well as physiological signals like heart rate, respiration and body movement.
- Feature Extraction: Relevant features such as heart rate variability, respiration patterns, movement intensity and environmental stability indicators are extracted from each data segment. These features represent meaningful sleep characteristics for machine learning analysis.
- Data Transmission: The gathered data is safely sent to an edge device for processing using wireless communication protocols (Wi-Fi/Bluetooth).
- Data Pre-processing: Preprocessing techniques will be applied for noise reduction and feature extraction. AI models, including machine learning and deep learning, classify sleep stages, evaluate sleep quality and detect irregularities in real time.
- Analysis: A trained Random Forest model which is pretrained is used to predict the sleep quality by taking the user data. For every incoming data, the model produces a predicted sleep stage and class probabilities [5], [10], [12].
- AI-Based Sleep Stage Classification: The extracted features are input to trained machine learning or deep learning models deployed in the cloud. The models classify sleep stages (awake, light sleep, deep sleep and REM) and compute an overall sleep quality score.

C. Mathematical Modelling and Notation

Let the sleep monitoring system collect multimodal data from IoT sensors over a sleep duration T. The collected dataset is represented as:

$$\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$$

where each feature vector $\mathbf{x}_i \in \mathbb{R}^d$ contains physiological and environmental parameters:

$$\mathbf{x}_i = [HR_i, RR_i, MV_i, Temp_i, Hum_i, Noise_i]$$

HR– Heart rate,

RR – Respiration rate,

MV – Body movement,

Temp – Temperature,

Hum – Humidity,

Noise – Ambient sound level.

Data Preprocessing

Noise reduction is performed using a filtering function $f(\cdot)$:

$$\mathbf{X}' = f(\mathbf{X}) \quad (1)$$

where $\mathbf{X}'$ denotes the cleaned and normalized dataset [5], [7].

Feature Extraction

Extracted features are computed as:

$$\mathbf{F} = g(\mathbf{X}') \quad (2)$$

where $g(\cdot)$ represents feature extraction methods such as heart rate variability, statistical moments and movement intensity [5].

Sleep Stage Classification

The sleep stage prediction problem is modelled as a multi-class classification task:

$$\hat{y} = h(\mathbf{F}) \quad (3)$$

where,
$h(\cdot)$ is the trained machine learning model (Random Forest / Deep Learning), $\hat{y} \in \{Awake, Light, Deep, REM\}$.

For a Random Forest classifier with M decision trees:

$$\hat{y} = \arg\max_c \sum_{m=1}^{M} \mathbb{I}(h_m(\boldsymbol{F}) = c) \quad (4)$$

where $\mathbb{I}(\cdot)$ is the indicator function [5], [7].

Sleep Quality Score Estimation

The overall sleep quality score Q is calculated as:

$$Q = \sum_{k=1}^{K} w_k \cdot S_k \quad (5)$$

Where,
S_k represents the normalized sleep metrics (sleep duration, efficiency and REM ratio), w_k denotes the importance weight of each metric [2].

IV. System Architecture

The proposed AI-IoT Sleep Monitoring and Enhancement System is designed with a layered architecture for scalability and adaptability. It consists of three layers: IoT Sensing Layer, AI Processing Layer and output layer. By combining these three layers facilitate both continuous monitoring and real time improvement of sleep quality [13].

A. IoT Sensing Layer

The IoT sensing layer forms the foundation of the system and is responsible for continuous data acquisition. It consists of wearable sensors for monitoring physiological parameters such as heart rate, respiration rate, blood oxygen saturation and body movement. In addition, non-wearable environmental sensors are deployed to measure ambient conditions including temperature, humidity, light intensity and noise levels. These multimodal sensor inputs provide the comprehensive representation of the factors influencing sleep quality [1], [8].

All sensors operate in a low-power mode to ensure prolonged usage and minimal user discomfort. The collected data are time-synchronized and transmitted securely using wireless communication protocols such as Bluetooth Low

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Energy (BLE) and Wi-Fi, ensuring reliable data transfer with minimal latency [6], [7].

B. AI Processing Layer

The AI processing layer is responsible for all the data preprocessing, feature extraction and intelligent analysis. Initially, raw sensor data are filtered to remove noise followed by normalization and synchronization of multimodal signals. Feature extraction techniques derive meaningful attributes such as heart rate variability, movement intensity and environmental stability indices. Machine learning models, including Random Forest and deep learning architectures, analyse the extracted features to classify sleep stages and evaluate sleep quality [5], [10]. To support real-time analysis and scalability, the system employs a hybrid edge–cloud computing approach. Edge devices handle preliminary processing to reduce latency, while cloud-based AI models perform advanced analytics, long-term trend analysis and adaptive learning. This architecture enables efficient handling of continuous health data streams and supports personalized model updates [12], [13].

C. Output

This system uses wireless communication methods to collect sleep-related data from the user in real time. It allows users to view their sleep results easily on a mobile device. Along with showing daily sleep data, the system also tracks long-term sleep patterns and provides personalized recommendations to help improve sleep quality [8], [11].

V. METHODOLOGIES

The system combines IoT-based sensing and AI-driven analytics into a multi-stage process. This process enables continuous data collection, accurate analysis to enhance sleep quality in real-time.

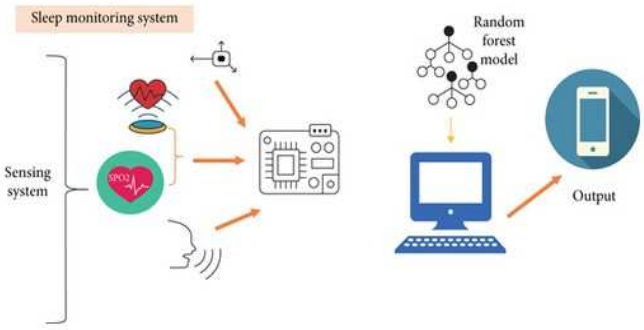


Fig. 2. Block diagram of the sleep monitoring system

Figure 2 illustrates the workflow of a sleep monitoring system that integrates sensing technology with machine learning for accurate sleep analysis. The sensing system collects physiological signals such as heart rate, blood oxygen level and sound patterns (like breathing or snoring) through various sensors. These signals are transmitted to a processing unit, which organizes and preprocesses the data.

The processed data is then sent to a computer system, where a Random Forest model analyses it to determine sleep stages or detect abnormalities in sleep behaviour. Finally, the analysed results are displayed on a mobile device on their sleep quality and related parameters Fig(2). Overall, this architecture demonstrates a smart, data-driven approach for continuous and non-intrusive sleep monitoring using sensor integration and machine learning algorithms.

A. Data Collection

Wearable and ambient sensors enabled with IoT technology are used to collect both physiological and environmental variables related to sleep. Physiological signals include heart rate, respiratory rate and body movement. Environmental signals include temperature, humidity, light intensity and sound pressure levels. The data collected by the sensors are sent to a processing unit by the secure wireless protocols.

B. Data Preprocessing

Sensor-obtained data contains some disturbances in the signals. Data preprocessing methods are used to improve the data quality, including:

- Noise Reduction: Filtering techniques that helps to reduce the disturbances in the signals (noise) while preserving sleep quality.
- Feature Extraction / Derivation: Creating meaningful features from datasets such as heart-rate variability and movement intensity.
- Data Synchronization: Combining multi-signals from different sensors together to perform accurate data analysis [10].

C. Feature Engineering and Selection

As a way to reduce the dimensionality and improve computational efficiency for AI models, relevant physiological and environmental features are identified from datasets. Statistical methods, correlation analysis, and domain knowledge are utilized to determine the most discriminative features used in analysing sleep quality.

D. AI-Based Evaluation

Sleep stage classification (Light, Deep, REM) and overall sleep quality are determined using machine learning and deep learning models. The conceptual approach includes:

- Supervised Learning Models: Baseline classification is accomplished using traditional algorithms such as Random Forest and Support Vector Machines (SVM).
- Deep Learning Models: Advanced models such as Long Short-Term Memory (LSTM) networks and Convolutional Neural Networks (CNN) are applied to capture sequential and temporal patterns in physiological signals.
- Personalization: Adaptive algorithms continuously refine their predictions by learning from an individual's long-term sleep characteristics [5], [10], [11], [8].

E. Evaluation and Validation

The validation of the proposed system is carried out using well-recognized benchmark datasets along with data collected from real-time sensing devices. To quantify the system's performance, commonly adopted evaluation indicators namely

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accuracy, precision, recall and F1-score are utilized. In addition, a comparative investigation with existing solutions reveals that the developed framework delivers notable enhancements, particularly with respect to predictive accuracy for larger deployments.

VI. Results and Discussion

The proposed AI-powered IoT sleep monitoring framework was evaluated using both benchmark datasets and real-time sensor data collected from a controlled environment Fig (3). The results demonstrate the system's effectiveness in accurate sleep stage classification and user-specific personalization.

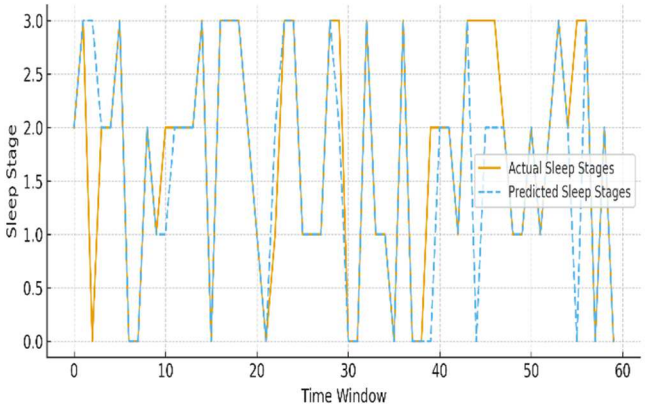


Fig. 3. Random Forest Predictions

A. Sleep Stage Classification Accuracy

- The ML models such as Random Forest achieved an accuracy level between 82%–86% Fig (3).
- Deep learning models Outperformed traditional models, achieving accuracy levels above 90%.
- Personalized models further improved classification accuracy by 3–5% due to individual-specific adaptations.

B. Performance Metrics

The evaluation employed standard data science metrics. Results indicated:

- **Precision:** 91.2%
- **Recall:** 89.7%
- **F1-Score**: 90.4%

$$\text{Precision} = \frac{TP}{TP + FP} \tag{6}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{7}$$

$$\text{F1-score} = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall} \tag{8}$$

C. Comparative Analysis

In comparison with previous studies developed frameworks, the proposed model exhibits enhanced flexibility and scalability. Unlike earlier approaches that were limited to offline evaluation, this system integrates IoT-enabled real-time data acquisition with adaptive artificial intelligence algorithms. This combination facilitates continuous observation ensuring a more responsive and efficient monitoring process.

D. Discussion

The findings emphasize the advantage of merging IoT-based sensing technologies with intelligent data analytics [14]. Although deep learning methods contribute to better identification of sleep stages, the inclusion of personalized calibration is crucial for maintaining long-term precision and improving user experience. Future developments may aim to broaden the range of sensing inputs, enhance wearable comfort and establish seamless compatibility across multiple platforms to promote broader usability and deployment.

VII. Conclusion

An intelligent framework is proposed for continuous monitoring and enhancement of sleep quality. By combining data from wearable and environmental sensors with machine learning algorithms, the proposed system achieved precise classification of sleep stages along with adaptive user personalization. The experimental evaluation confirmed that deep learning architectures consistently surpassed conventional approaches in detecting and classifying sleep stages with higher accuracy [15]. Moreover, this strategy improved the user engagement over time. In conclusion, the future research this model can be able to take multimodal physiological data and combine it directly with various algorithms across platforms and extend the usability for larger deployment in the healthcare and personalized wellness domains. Edge and fog computing paradigms offer lower latency processing by bringing computation closer to IoT endpoints. Aazam and Huh discussed the use of fog-based gateways to handle data preprocessing and analytics near the data source, reducing dependency on cloud servers for every computation. In sleep quality monitoring, such edge strategies enable near real-time inference and rapid feedback to users, improving timeliness and reducing communication overhead.

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